

1	Course Name:	Software Project Management																																																																																																																																																																																																		
	Course Code:	BMSE3153																																																																																																																																																																																																		
	Course Classification:	Major (core)																																																																																																																																																																																																		
2	Synopsis:	This course introduces students to the essential aspects of software and project management which includes the application of various proven software project management techniques, methods and tools such as project planning and estimation, project monitoring and control, software process models and project metrics for cost effective, reliable and quality software.																																																																																																																																																																																																		
3	Name(s) of Academic Staff:	1	Refer to timetable																																																																																																																																																																																																	
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4	Semester and Year offered:	Year Offered			Semester		Remarks: Refer to Programme Structure																																																																																																																																																																																													
5	Credit Value:	3																																																																																																																																																																																																		
6	Pre-requisite/ co-requisite (if any):	BMSE1003 Software Engineering OR BMSE2163 Software Engineering OR BACS2163 Software Engineering (for Bachelor of Software Engineering (Honours)) Nil (for all other programmes)																																																																																																																																																																																																		
7	Course Learning Outcomes (CLO) 	CLO1	Analyse the software project management techniques, methods and processes for project management. (C4, PLO2)																																																																																																																																																																																																	
		CLO2	Assume responsibility to practice communication skills for project management. (A3, PLO5)																																																																																																																																																																																																	
		CLO3	Behave according to the code of ethics and professional conduct for project management. (A3, PLO11)																																																																																																																																																																																																	
8	Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods																																																																																																																																																																																																			
<table border="1"> <thead> <tr> <th rowspan="2">Course Learning Outcomes</th> <th colspan="11">Programme Learning Outcomes (PLO)</th> <th rowspan="2">Teaching Methods</th> <th rowspan="2">Assessment Methods</th> </tr> <tr> <th>PLO 1</th> <th>PLO 2</th> <th>PLO 3</th> <th>PLO 4</th> <th>PLO 5</th> <th>PLO 6</th> <th>PLO 7</th> <th>PLO 8</th> <th>PLO 9</th> <th>PLO 10</th> <th>PLO 11</th> </tr> </thead> <tbody> <tr> <td>CLO1</td> <td></td> <td>v</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>L,T,P,NF2F</td> <td>Test, examination</td> </tr> <tr> <td>CLO2</td> <td></td> <td></td> <td></td> <td></td> <td>v</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>L,T,P,NF2F</td> <td>Assignment</td> </tr> <tr> <td>CLO3</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>v</td> <td></td> <td>L,T,P,NF2F</td> <td>Assignment</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td rowspan="3">Mapping with MQF Cluster of Learning Outcomes</td> <td></td> <td>C2</td> <td></td> <td></td> <td>C3C</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>C5</td> <td></td> <td rowspan="3"></td> <td rowspan="3"></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>													Course Learning Outcomes	Programme Learning Outcomes (PLO)											Teaching Methods	Assessment Methods	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	CLO1		v											L,T,P,NF2F	Test, examination	CLO2					v								L,T,P,NF2F	Assignment	CLO3											v		L,T,P,NF2F	Assignment																																																																												Mapping with MQF Cluster of Learning Outcomes		C2			C3C						C5																											
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Indicate the primary causal link between the CLO and PLO by ticking 'v' in the appropriate box. C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C = Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Entrepreneurial Skills, C5 = Ethics & Professionalism																																																																																																																																																																																																				
9	Transferable Skills (if applicable) <i>(Skills learned in the course of study which can be useful and utilized in other settings)</i> <table border="1"> <tr> <td>1</td> <td>Communication Skills</td> </tr> <tr> <td>2</td> <td>Ethics and Professionalism</td> </tr> <tr> <td>3</td> <td>Cognitive skills</td> </tr> </table> Open-ended response (if any)												1	Communication Skills	2	Ethics and Professionalism	3	Cognitive skills																																																																																																																																																																																		
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10 Distribution of Student Learning Time (SLT)

Note: This SLT calculation is designed for home grown programme only.

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**										Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)		
			Physical				Online/ Technology-mediated (Synchronous)						
			L	T	P	O	L	T	P	O			
1	Management and Project Management Concept • Project life cycle, needs identification, RFP and proposals • Project appraisal and evaluation • Attributes and knowledge of a project manager • Human resource management spectrum and Team Development • Causes of project failures and critical success factors • Generic project process model and its attributes	1,2,3	10	5	5	-						5	
2	Project Planning, Control and Process Models • Project planning (PERT and Gantt chart) and estimation • Project monitoring process and control • Selection of process models	1,2,3	2	1	5	-						1	
3	Quality Management and Assurance • Quality management and costs • Quality concept and principle activities • Quality assurance and standards • Process and product quality relationship • Quality planning and quality controls • Techniques to help enhance software quality • The ISO9001 standards framework	1,2,3	4	2	1	-						2	
4	Software Metrics • Metrics for software quality • Function oriented and line of code metrics • Attributes of good metrics and implementation of metrics	1,2,3	2	1	1	-						1	
5	Software Process Improvement • The Software Process Improvement (SPI) • Importance of SPI • The CMMI process improvement framework	1,2,3	2	1	1	-						1	
6	Software Configuration Management •Software Configuration Management (SCM) concepts (baselines, SCM items, etc) • SCM process • Version control and change control • Configuration audit and reporting	1	2	1	-	-						1	

7	Risk Management • Risk analysis and management • Types of risks • Risk Identification • Risk Projection • Risk Refinement • RMMM and safety risks and hazards	1,2,3	2	1	1	-						1
8	Software Dependability & Critical Systems • Dimensions of system dependability • Types of critical systems • Systems availability and reliability • Approaches to improve system reliability • Safety and ways of improving system safety • Security and its damages • Approaches to secure system security	1	4	2	-	-						2
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
SUB-TOTAL SLT:												70
Continous Assessment		%	Face-to-Face (F2F)		NF2F							
			Physical		Online/ Technology-mediated (Synchronous)		Independent Learning for Assessment (Asynchronous)					
1	Test	10	1				10					
2	Assignment	40	7				20					
3												
4												
5												
SUB-TOTAL SLT:												38
Final Assessment		%	Face-to-Face (F2F)		NF2F							
			Physical		Online/ Technology-mediated (Synchronous)		Independent Learning for Assessment (Asynchronous)					
1	Examination	50	2				10					
2												
3												
4												
5												
SUB-TOTAL SLT:												12

SLT for Assessment:		50
GRAND TOTAL SLT:		120
A	% SLT for F2F Physical Component: $\frac{[Total\ F2F\ Physical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$	55.00
B	% SLT for Online & Independent Learning Component: $\frac{[(Total\ F2F\ Online + Total\ Independent\ Learning)]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$	45.00
C	% SLT for All Practical Component: $\frac{[%\ F2F\ Physical\ Practical + \%\ F2F\ Online\ Practical]}{[Total\ F2F\ Physical\ Practical + Total\ F2F\ Online\ Practical + Total\ Independent\ Learning]} \times 100]$	11.67
C1	% SLT for F2F Physical Practical Component: $\frac{[Total\ F2F\ Physical\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$	11.67
C2	% SLT for F2F Online Practical Component: $\frac{[Total\ F2F\ Online\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$	0.00

Please tick (v) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

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Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requirements imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	Nil
12	References (include required and further readings, and should be the most current)	Main references supporting the course 1. Kloppenborg T., Vittal S. A., & Wells, K. (2022). <i>Contemporary Project Management</i> (5th ed). Cengage Learning. https://tarcez.tarc.edu.my/login?url=https://resolver.vitalsource.com/9780357715826 2. Tsui, F., Karam, O., & Bernal, B. (2023). <i>Essentials of Software Engineering</i> (5th ed.). Jones & Bartlett Learning. https://tarcez.tarc.edu.my/login?url=https://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=3135670
13	Other additional information (if applicable)	Nil

Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.